

Unifying Special Relativity and Quantum Mechanics

Lecture 14 — The Massive Spin-1 Field and its Massless Limit

(Proca, Ghosts, the Stückelberg Decomposition,
Conserved Currents, and the Photon Propagator)

Typed version of the handwritten notes, with added explanations in blue boxes

How to read this document

Conventions. Exactly as in Lectures 1–13: everything in the normal text and in the **yellow / pink** boxes is a faithful transcription of the handwritten notes (the coloured boxes reproduce the results that were framed or highlighted in the original — yellow boxes for the yellow-highlighted results, pink boxes for the pink/red-framed ones). Everything inside a **blue box like this one** is extra material that I added: derivations spelled out, terminology, corrections of transcription slips, and connections to earlier lectures.

Pagination of this scan. These sheets continue, without interruption, the chapter that Lecture 13 opened (the seventh restart of the page numbering). They are therefore marked [p. 21*****]–[p. 40*****], with **seven stars**, exactly as in Lecture 13.

Two pagination anomalies in this scan.

- (a) **Overlap.** The first two sheets, [p. 21*****] and [p. 22*****], are a *repeat*: they are the last two pages of the Lecture 13 scan. They are reproduced here in condensed form (Section 1) so that the Lagrangian ansatz this lecture starts from is on the page.
- (b) **Missing pages.** Sheets [p. 24*****] and [p. 25*****] are *absent from the scan*. Page 26***** opens mid-expression, with the fragment $M^2 V_{T,\mu}(\partial^\mu \phi)$ left over from the previous sheet. The blue box in Section 3 reconstructs what the missing pages must have contained — the substitution of the Helmholtz decomposition into the Lagrangian — so that the argument can be followed without them.

Where we left off. Lecture 13 rebuilt QFT on the Lagrangian formalism (axioms, Noether, canonical quantization) and classified interactions by mass dimension. Its last page wrote down the most general quadratic ansatz for a massive spin-1 field, with two undetermined coefficients α, β . This lecture fixes them, discovers that the answer is forced twice over (once by the constraint, once by the absence of ghosts), and then spends the rest of its length on the question that has been hanging since Lecture 11: *can the massless limit be taken at all?*

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Part I — The free massive spin-1 field

1 Recap: mass dimensions and the Lagrangian ansatz [p. 21*****–22*****]

These two pages repeat the end of Lecture 13

They are transcribed in full in the Lecture 13 document; here is the condensed content, since everything that follows depends on it.

Marginal interactions (“renormalizable”): operators of mass dimension exactly 4. Example $\mathcal{L}_{\text{int}} = g V_\mu(x) \bar{\Psi}(x) \gamma^\mu \Psi(x)$, with $[\Psi] = [M]^{3/2}$ and $[V^\mu] = [M]$; the coefficient is dimensionless.

Irrelevant interactions (“non-renormalizable”): mass dimension larger than 4. Example $\mathcal{L}_{\text{int}} = \kappa F_{\mu\nu}(x) \bar{\Psi}(x) \sigma^{\mu\nu} \Psi(x)$, of dimension $5 > 4$, so that $[\kappa] = [M]^{-1}$ — a negative mass dimension.

The massive spin-1 field — the free theory

Consider the field $V^\mu(x)$. As a free field, it satisfies the equations

$$(\square + M^2)V^\mu(x) = 0, \quad \partial_\mu V^\mu(x) = 0.$$

We look for a Lagrangian description.

$\hookrightarrow$ For the free theory, we look for a Lagrangian with terms quadratic in the fields and at most quadratic in derivatives. We take into account that $[V^\mu] = [M]$.

We write

$$\mathcal{L} = -\frac{\alpha}{2}(\partial_\mu V_\nu)(\partial^\mu V^\nu) - \frac{\beta}{2}(\partial_\mu V_\nu)(\partial^\nu V^\mu) + \frac{1}{2}M^2 V_\mu V^\mu$$

2 The equation of motion fixes $\beta = -\alpha$ [p. 23*****]

We compute the equation of motion, $\frac{\partial \mathcal{L}}{\partial V^\mu} = \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu V^\nu)}$, from

$$\frac{\partial \mathcal{L}}{\partial V^\nu} = \frac{1}{2}M^2(2V_\nu) = M^2 V_\nu, \quad \frac{\partial \mathcal{L}}{\partial(\partial_\mu V^\nu)} = -\alpha(\partial^\mu V_\nu) - \beta(\partial_\nu V^\mu)$$

So that

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial V^\nu} = \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu V^\nu)} &\rightsquigarrow M^2 V_\nu = -\partial_\mu (\alpha \partial^\mu V_\nu + \beta \partial_\nu V^\mu) \\ &= -\alpha \square V_\nu - \beta \partial_\nu (\partial_\mu V^\mu) \end{aligned}$$

That is

$$\alpha \square V_\mu + \beta \partial_\mu (\partial_\nu V^\nu) + M^2 V_\mu = 0 \quad (\star)$$

We take the four-divergence:

$$\alpha \square (\partial^\mu V_\mu) + \beta \square (\partial_\mu V^\mu) + M^2 (\partial_\mu V^\mu) = 0$$

$$(\alpha + \beta)\square(\partial_\mu V^\mu) + M^2(\partial_\mu V^\mu) = 0$$

If we take $(\alpha + \beta) = 0$, that is $\beta = -\alpha$, we get

$$M^2(\partial_\mu V^\mu) = 0 \quad \rightsquigarrow \quad (\partial_\mu V^\mu) = 0$$

That is precisely the constraint we wanted. We now come back to the E.o.M. $(\star)$ and set $(\partial_\mu V^\mu) = 0$.

Added explanation (not in the original notes)

Why the four-divergence is the right thing to take. The Lagrangian gives *one* equation, $(\star)$, but the free massive spin-1 field must satisfy *two*: Klein–Gordon and the transversality constraint. Contracting $(\star)$ with ∂^μ produces a scalar equation for the single quantity $\partial_\mu V^\mu$, and asking that this equation force $\partial_\mu V^\mu = 0$ is what pins down the ratio β/α .

What goes wrong if $\alpha + \beta \neq 0$. The divergence equation becomes $[\square + M^2/(\alpha + \beta)](\partial_\mu V^\mu) = 0$, i.e. $\partial_\mu V^\mu$ is a *propagating scalar field* of mass $M^2/(\alpha + \beta)$ rather than being zero. The theory then describes a spin-1 particle *plus* an unwanted spin-0 particle — and, as [p. 26*****] shows, that extra scalar is a ghost. Only $\alpha + \beta = 0$ removes it.

Completing the derivation. With $\beta = -\alpha$ and $\partial_\mu V^\mu = 0$, $(\star)$ collapses to $\alpha\square V_\mu + M^2 V_\mu = 0$, which is the Klein–Gordon equation $(\square + M^2)V_\mu = 0$ provided $\alpha = 1$. That normalization is a choice of units for the field, and with it the two derivative terms assemble into

$$-\frac{1}{2}(\partial_\mu V_\nu)(\partial^\mu V^\nu - \partial^\nu V^\mu) = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu},$$

giving the **Proca Lagrangian** $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2 V_\mu V^\mu$ anticipated in Lecture 10.

3 The missing pages, and the ghost argument [p. 26*****]

Reconstruction of the missing sheets 24*****–25*****

Page 26***** begins with the isolated fragment $M^2 V_{T,\mu}(\partial^\mu \phi)$, which is the tail of a longer expression. What the two absent sheets must have done is substitute the **Helmholtz decomposition** of Lecture 10 ([p. 25****]),

$$V^\mu = V_T^\mu + \partial^\mu \phi, \quad \partial_\mu V_T^\mu = 0,$$

into the general ansatz and expand. The mass term alone gives

$$\frac{1}{2}M^2 V_\mu V^\mu = \frac{1}{2}M^2 V_{T,\mu} V_T^\mu + \frac{1}{2}M^2 (\partial_\mu \phi)(\partial^\mu \phi) + \underbrace{M^2 V_{T,\mu}(\partial^\mu \phi)}_{\text{the fragment atop p. 26*****}},$$

and the two derivative terms produce, among others, the pieces proportional to $(\alpha + \beta)$ that the page immediately goes on to discuss. So nothing conceptual is lost: the missing sheets are the algebra of that substitution.

... $M^2 V_{T,\mu}(\partial^\mu \phi)$

What forces the condition $\alpha + \beta = 0$? Consider the terms proportional to $(\alpha + \beta)$:

$$\frac{1}{2}(\alpha + \beta)(\partial_\mu \partial_\nu \phi)(\partial^\mu \partial^\nu \phi) \quad \text{and} \quad -(\alpha + \beta)(\partial^\mu V_T^\nu)(\partial_\mu \partial_\nu \phi)$$

$\hookrightarrow$ Quadratic in the fields but with second-order derivatives in the field ϕ . These terms are problematic in QFT because they indicate the presence of physical ghosts (and typically lead to violation of unitarity and causality).

Consequently, the condition $\alpha + \beta = 0$ can be also derived by imposing the absence of physical ghosts.

Let us now impose the condition $\alpha + \beta = 0$ and $\alpha = 1$ in the Lagrangian written in terms of V_T^μ and ϕ :

$$\mathcal{L} = -\frac{1}{2} \left[(\partial_\mu V_{T,\nu}) (\partial^\mu V_T^\nu) - (\partial_\mu V_{T,\nu}) (\partial^\nu V_T^\mu) \right] + \frac{1}{2} M^2 V_{T,\mu} V_T^\mu + \frac{1}{2} M^2 (\partial_\mu \phi) (\partial^\mu \phi) + M^2 V_{T,\mu} (\partial^\mu \phi)$$

We introduce a rescaling of the field ϕ such that ...

Added explanation (not in the original notes)

What a ghost is, and why higher derivatives produce one. A *ghost* is a state of negative norm, or equivalently a field whose kinetic term comes with the wrong sign. Its presence is fatal: negative-norm states give negative probabilities, so unitarity fails.

The link to higher derivatives is **Ostrogradsky's theorem**, mentioned in Lecture 13 when the action was restricted to first derivatives. A Lagrangian containing $\partial^2 \phi$ has a Hamiltonian that is linear in one of its momenta and therefore unbounded below — the energy can be made arbitrarily negative. On quantization this shows up as an extra pole in the propagator with the opposite sign of residue: a negative-norm state.

To see it concretely, a term $\frac{c}{2}(\Box\phi)^2$ added to a scalar kinetic term gives a propagator

$$\frac{1}{p^2 - c p^4} = \frac{1}{p^2} - \frac{1}{p^2 - 1/c},$$

i.e. the healthy massless pole *minus* a second pole — the relative minus sign is the ghost.

Why this is a genuinely independent derivation. The route on [p. 23****] was: *demand the constraint* $\partial_\mu V^\mu = 0$, get $\beta = -\alpha$. The route here is: *demand no ghosts*, get $\beta = -\alpha$. Two different physical requirements, one answer. That is worth pausing on — it means the Proca Lagrangian is not a convenient choice but the only consistent one, and it explains why the notes emphasize the point.

Reading the two $(\alpha + \beta)$ terms. Both involve $\partial_\mu \partial_\nu \phi$, second derivatives of the scalar. Note that they arise *only* from the longitudinal piece of the decomposition: the transverse part V_T never generates them. So the scalar ϕ — which will shortly be recognized as the longitudinal polarization — is exactly where the danger lives.

4 Canonical normalization and the Proca Lagrangian [p. 27****]

... $\phi \rightarrow \frac{1}{M} \phi$:

$$\mathcal{L} = -\frac{1}{2} \left[(\partial_\mu V_{T,\nu}) (\partial^\mu V_T^\nu) - (\partial_\mu V_{T,\nu}) (\partial^\nu V_T^\mu) \right] + \frac{1}{2} M^2 V_{T,\mu} V_T^\mu + \underbrace{\frac{1}{2} (\partial_\mu \phi) (\partial^\mu \phi)} + M V_{T,\mu} (\partial^\mu \phi)$$

Now the field ϕ is “canonically normalized” and it has mass-dimension 1.

The Helmholtz decomposition now reads

$$V^\mu = V_T^\mu + \frac{1}{M} (\partial^\mu \phi), \quad \partial_\mu V_T^\mu = 0$$

We further notice that

$$F^{\mu\nu} = \partial^\mu V^\nu - \partial^\nu V^\mu = \partial^\mu \left(V_T^\nu + \frac{1}{M} \partial^\nu \phi \right) - \partial^\nu \left(V_T^\mu + \frac{1}{M} \partial^\mu \phi \right)$$

$$= \partial^\mu V_T^\nu + \frac{1}{M} \cancel{\partial^\mu \partial^\nu \phi} - \partial^\nu V_T^\mu - \frac{1}{M} \cancel{\partial^\nu \partial^\mu \phi} = \partial^\mu V_T^\nu - \partial^\nu V_T^\mu$$

So that the combination $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is

$$-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = -\frac{1}{2}(\partial_\mu V_{T,\nu})(\partial^\mu V_T^\nu - \partial^\nu V_T^\mu) = -\frac{1}{2}(\partial_\mu V_{T,\nu})(\partial^\mu V_T^\nu) + \frac{1}{2}(\partial_\mu V_{T,\nu})(\partial^\nu V_T^\mu)$$

We get the Lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2 V_{T,\mu}V_T^\mu + \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) + M V_{T,\mu}(\partial^\mu \phi)$$

with $V^\mu \equiv V_T^\mu + \frac{1}{M}(\partial^\mu \phi)$.

Added explanation (not in the original notes)

Why the rescaling was needed. Before it, the scalar kinetic term was $\frac{1}{2}M^2(\partial\phi)^2$ — non-standard, and it forces $[\phi] = [M]^0$, i.e. a dimensionless field. After $\phi \rightarrow \phi/M$ the kinetic term is the textbook $\frac{1}{2}(\partial\phi)^2$ and $[\phi] = [M]$, matching every other scalar in the course. This is what “canonically normalized” means. Note that the price is a factor $1/M$ in the decomposition — and that factor is precisely what will blow up in the massless limit.

The cancellation in $F^{\mu\nu}$ is the key structural fact. Because $\partial^\mu \partial^\nu \phi$ is symmetric in $\mu \leftrightarrow \nu$ while $F^{\mu\nu}$ is antisymmetric, the scalar drops out of the field strength entirely. So $F^{\mu\nu}$ knows only about V_T , and the Maxwell term is blind to ϕ . Everything that couples ϕ to V_T sits in the mass sector.

This is the Stückelberg trick. What has been done here has a name: introducing a scalar ϕ so that a massive vector Lagrangian can be written in a form where the mass term does not obstruct gauge invariance. In the Stückelberg language the combined Lagrangian is invariant under

$$V_T^\mu \rightarrow V_T^\mu + \partial^\mu \alpha, \quad \phi \rightarrow \phi - M\alpha,$$

and choosing $\phi = 0$ (“unitary gauge”) recovers the Proca form. The formulation is the standard tool for studying massive vectors at high energy, and it is the free-field ancestor of the Higgs mechanism, where ϕ becomes the would-be Goldstone boson eaten by the gauge field.

5 The massless limit and the DOF counting [p. 28*****-29*****]

This Lagrangian is equivalent to the Proca Lagrangian and propagates only **3 DOF**.

It is instructive to take the massless limit, $M \rightarrow 0$:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi)$$

$$\begin{aligned} F^{\mu\nu} &= \partial^\mu V_T^\nu - \partial^\nu V_T^\mu \\ \partial_\mu V_T^\mu &= 0 \end{aligned}$$

Free Maxwell Lagrangian: it propagates only **2 DOF**

Massless scalar field, 1 DOF

The condition $\partial_\mu V_T^\mu = 0$ eliminates 1 DOF. In addition, the Lagrangian $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is invariant under the gauge transformation $V_T^\mu \rightarrow V_T^\mu + \partial^\mu \alpha$ that preserves $\partial_\mu V_T^\mu = 0$ if $\square\alpha = 0$. This “residual” gauge freedom further eliminates 1 DOF.

The decomposition $V^\mu = V_T^\mu + \frac{1}{M}(\partial^\mu \phi)$ allows for a clear separation of DOFs in the massless limit. In fact, let us write

$$\begin{aligned}
 V^\mu(x) &= V_T^\mu(x) + \frac{1}{M}(\partial^\mu \phi) \\
 &\quad \hookrightarrow \text{since } \phi \text{ is a real massless field} \\
 &= V_T^\mu(x) + \frac{1}{M} \partial^\mu \left[\int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left(e^{-ip \cdot x} a(\vec{p}) + e^{ip \cdot x} a^\dagger(\vec{p}) \right) \right] \\
 &= V_T^\mu(x) + \frac{1}{M} \int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left(-ip^\mu e^{-ip \cdot x} a(\vec{p}) + ip^\mu e^{ip \cdot x} a^\dagger(\vec{p}) \right) \\
 V^\mu &= \underbrace{V_T^\mu}_{\text{two transverse polarizations}} + \underbrace{\int \frac{d^3 \vec{p}}{(2\pi)^3 \sqrt{2E_p}} \left(\frac{-ip^\mu}{M} e^{-ip \cdot x} a(\vec{p}) + \frac{ip^\mu}{M} e^{ip \cdot x} a^\dagger(\vec{p}) \right)}_{\text{longitudinal polarization in the ultra-relativistic limit (massless limit)}}
 \end{aligned}$$

We summarize these findings by saying that in the massless limit

$$V^\mu = \underbrace{V_T^\mu}_{\text{two transverse DOF}} + \underbrace{\frac{1}{M}(\partial^\mu \phi)}_{\text{1 longitudinal DOF}}, \quad \partial_\mu V^\mu = 0$$

Added explanation (not in the original notes)

The counting, carefully. A four-vector has 4 components. In the massive theory the constraint $\partial_\mu V^\mu = 0$ removes one, leaving $3 = 2s + 1$ for $s = 1$: two transverse plus one longitudinal. In the massless theory the same constraint removes one, and then *gauge invariance* removes a second, leaving 2 — the two helicities of the photon. The extra reduction is available only at $M = 0$, because the mass term $\frac{1}{2}M^2 V_\mu V^\mu$ is not gauge invariant.

Why the residual gauge parameter must satisfy $\square \alpha = 0$. Under $V_T^\mu \rightarrow V_T^\mu + \partial^\mu \alpha$ the divergence changes by $\square \alpha$. Preserving $\partial_\mu V_T^\mu = 0$ therefore requires $\square \alpha = 0$ — one function's worth of freedom remaining after Lorenz gauge has been imposed, hence exactly one more degree of freedom removed.

Why $\frac{1}{M} \partial^\mu \phi$ is the longitudinal polarization. Comparing with the mode expansion of Lecture 9, a polarization vector ε^μ multiplies $e^{-ip \cdot x} a(\vec{p})$. The bracket above shows the longitudinal piece carries $\varepsilon_L^\mu = -ip^\mu/M$, i.e. $\varepsilon_L^\mu \propto p^\mu/M$ — exactly the high-energy form of the longitudinal polarization vector quoted on [p. 32*****]. This is the precise sense in which the scalar ϕ “is” the longitudinal mode.

The trouble, stated in advance. $\varepsilon_L^\mu \sim p^\mu/M$ diverges as $M \rightarrow 0$. The free Lagrangian above hid this, because at $M = 0$ the two sectors simply decoupled into Maxwell plus a free scalar and nothing blew up. The divergence becomes visible only once *interactions* are switched on — which is what the rest of the lecture is about.

The DOF discontinuity. $3 \rightarrow 2 + 1$: the massless theory has 2 photon states, and the third does not vanish — it becomes a separate, decoupled massless scalar. The number of *spin-1* states jumps discontinuously from 3 to 2. This is the “ $3 \neq 2$ ” remark on [p. 32*****].

Part II — The interacting theory and the massless limit

6 The interacting theory [p. 30*****–31*****]

We consider the Lagrangian density

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu - V_\mu j^\mu$$

four-vector current with mass dimension $[M]^3$

Two typical situations:

i) $j^\mu(x)$ is a c-number external current

- An external current acts as an external source, and it does not evolve dynamically within the QFT.
- A prototypical example of external current is the electromagnetic current generated by a heavy nucleus. If the mass is very large compared to the typical energy scale of a process, we treat it as a classical source:

$$j^\mu(\vec{x}) = (Ze\delta(\vec{x}), \vec{0})$$

ii) $j^\mu(x)$ is an operator that depends on fields other than $V^\mu(x)$

In this case, additional terms describing the dynamics of the extra fields must be added.

Example: interactions with fermions described by a Dirac field.

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu + \underbrace{\bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi}_{\text{free part of } \mathcal{L} \text{ describing a free Dirac field}} \underbrace{-gV_\mu\bar{\Psi}\gamma^\mu\Psi - g_5V_\mu\bar{\Psi}\gamma^\mu\gamma_5\Psi}_{\text{interactions}}$$

Since $[\Psi] = [M]^{3/2}$ and $[V^\mu] = [M]$ it is not possible to write super-renormalizable interactions. These interactions we wrote are renormalizable: g and g_5 are dimensionless coupling constants.

Added explanation (not in the original notes)

Why $[j^\mu] = [M]^3$. The term $V_\mu j^\mu$ must have dimension 4, and $[V^\mu] = [M]$, so $[j^\mu] = [M]^3$. Consistently, the fermionic current $\bar{\Psi}\gamma^\mu\Psi$ has dimension $2 \times \frac{3}{2} = 3$ — which is exactly why g comes out dimensionless.

The two structures γ^μ and $\gamma^\mu\gamma_5$ are the *vector* and *axial-vector* currents. Both are dimension-3 and Lorentz-covariant, so on dimensional grounds alone both are allowed. They differ under parity: the vector current is a true four-vector, the axial one changes sign. The distinction will turn out to be decisive on [p. 37*****], where only the vector current is conserved.

Why no super-renormalizable interaction exists here. A super-renormalizable term needs dimension < 4 . With $[V] = 1$ and $[\Psi] = 3/2$, the smallest non-trivial combinations are V^2 (dimension 2, but that is the mass term, not an interaction), V^3 (dimension 3 — forbidden

by Lorentz invariance, since one cannot contract three vector indices without a derivative), and $\bar{\Psi}\Psi$ (dimension 3 — again a mass term). Nothing genuinely interacting sits below 4.

7 The massless limit: the equation of motion [p. 32*****–33*****]

We are especially interested in the massless limit of the massive spin-1 theory. We already know that this limit is tricky, since the longitudinal polarization goes like

$$\varepsilon^\mu(\vec{p}, \lambda = 0) = \frac{p^\mu}{M} \quad \text{and diverges as } M \rightarrow 0.$$

↔ From a physical point of view, the limit $M \rightarrow 0$ is not “smooth” because there is a discontinuity in the number of DOF going from the massive to the massless case ($3 \neq 2!$).

Let us investigate the E.o.M., considering the theory

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu - V_\mu j^\mu \quad \triangle$$

► We assume that j^μ does not depend on V^μ and/or $\partial_\nu V^\mu$ (we will relax this assumption later).

From $\frac{\partial \mathcal{L}}{\partial V^\mu} = \partial_\nu \frac{\partial \mathcal{L}}{\partial(\partial_\nu V^\mu)}$ we compute

$$\frac{\partial \mathcal{L}}{\partial V^\mu} = M^2 V_\mu - j_\mu$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial(\partial_\nu V_\mu)} &= \frac{\partial \mathcal{L}}{\partial F_{\rho\sigma}} \frac{\partial F_{\rho\sigma}}{\partial(\partial_\nu V_\mu)} = \left(-\frac{1}{2}F^{\rho\sigma}\right) \times \frac{\partial}{\partial(\partial_\nu V_\mu)} (\partial_\rho V_\sigma - \partial_\sigma V_\rho) \\ &= -\frac{1}{2}F^{\rho\sigma} \times (\delta_\rho^\nu \delta_\sigma^\mu - \delta_\sigma^\nu \delta_\rho^\mu) = -\frac{1}{2}(F^{\nu\mu} - F^{\mu\nu}) = -F^{\nu\mu} \end{aligned}$$

So that $M^2 V^\mu - j^\mu = -\partial_\nu (F^{\nu\mu})$:

$$\partial_\nu F^{\nu\mu} + M^2 V^\mu - j^\mu = 0$$

↔ If we take the four-divergence:

$$\underbrace{\partial_\mu \partial_\nu F^{\nu\mu}}_{=0} + M^2 (\partial_\mu V^\mu) - (\partial_\mu j^\mu) = 0$$

$$\text{The massless limit } M \rightarrow 0 \text{ implies } (\partial_\mu j^\mu) = 0$$

It seems possible to take the massless limit of a massive spin-1 theory if the latter interacts with a conserved current.

What happens to the issue of the longitudinal polarization that diverges in the limit $M \rightarrow 0$?

↔ To answer this question, we consider the decomposition $V^\mu = V_T^\mu + \frac{1}{M}\partial^\mu \phi$, $\partial_\mu V_T^\mu = 0$, since in the UR limit ϕ plays the role of the longitudinal polarization.

Added explanation (not in the original notes)

Reading the divergence equation. At finite M it says $\partial_\mu V^\mu = (\partial_\mu j^\mu)/M^2$: the constraint of the free theory is *deformed* by the source. As $M \rightarrow 0$ the left-hand side stays finite only if the numerator vanishes, so current conservation is not optional — it is a *consistency condition* for the massless limit to exist.

Why $\partial_\mu \partial_\nu F^{\nu\mu} = 0$. As in Lecture 10: a symmetric object $\partial_\mu \partial_\nu$ fully contracted with an antisymmetric $F^{\nu\mu}$ vanishes identically. This holds off-shell and independently of the dynamics.

The chain rule step, spelled out. $\mathcal{L}$ depends on $\partial_\nu V_\mu$ only through $F_{\rho\sigma}$, so one differentiates the Maxwell term with respect to F and then F with respect to ∂V . Using $-\frac{1}{4}F_{\rho\sigma}F^{\rho\sigma}$ one gets $\partial\mathcal{L}/\partial F_{\rho\sigma} = -\frac{1}{2}F^{\rho\sigma}$ (the factor 2 from the two identical F 's cancels one half), and the antisymmetry of F then converts $\frac{1}{2}(F^{\nu\mu} - F^{\mu\nu})$ into $F^{\nu\mu}$.

A caution flagged by the notes themselves. The red warning triangle marks the assumption that j^μ does not itself depend on V^μ . If it did — as happens in non-abelian gauge theory, where the current involves the gauge field — the variation would produce extra terms and the argument would need reworking.

8 Integration by parts saves the longitudinal mode [p. 33***–35*****]**

Starting from $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu - V_\mu j^\mu$ and substituting $V^\mu = V_T^\mu + \frac{1}{M}(\partial^\mu\phi)$:

$$\begin{aligned}\mathcal{L} = & -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_{T,\mu}V_T^\mu + \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) + M V_{T,\mu}(\partial^\mu\phi) \\ & - V_T^\mu j_\mu - \frac{1}{M}(\partial^\mu\phi)j_\mu\end{aligned}$$

↓ If we take the massless limit

$$\mathcal{L}(M \rightarrow 0) = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - V_T^\mu j_\mu + \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \underbrace{\frac{1}{M}(\partial^\mu\phi)j_\mu}$$

In the massless limit, we identify in the decomposition $V^\mu = V_T^\mu + \frac{1}{M}(\partial^\mu\phi)$ the term $\frac{1}{M}(\partial^\mu\phi)$ as the longitudinal DOF. In the $M \rightarrow 0$ limit, the interaction $-\frac{1}{M}(\partial^\mu\phi)j_\mu$ is problematic and it is indeed the one that involves the longitudinal DOF.

However, if the current is conserved, we can integrate by parts the term $-\frac{1}{M}(\partial^\mu\phi)j_\mu$, and get:

$$-\frac{1}{M}(\partial_\mu\phi)j^\mu \xrightarrow{\text{IBP}} +\frac{1}{M}\phi(\partial_\mu j^\mu) = 0$$

↔ If the current is conserved, this term vanishes on the E.o.M., and the longitudinal polarization does not pose any problem in the $M \rightarrow 0$ limit!

In summary:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu - V_\mu j^\mu$$

↓ admits $M \rightarrow 0$ limit if j^μ is conserved on the E.o.M.

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - V_\mu j^\mu, \quad \partial_\mu j^\mu = 0$$

Added explanation (not in the original notes)

Why the $1/M$ term is the whole problem. Every other term in the massless-limit Lagrangian is finite. Only $-\frac{1}{M}(\partial^\mu\phi)j_\mu$ carries an explicit $1/M$, and it is precisely the coupling of the *longitudinal* mode to matter. This is the Lagrangian-level counterpart of the statement that $\varepsilon_L^\mu \sim p^\mu/M$ diverges: the longitudinal polarization couples with divergent strength.

The resolution in one line. Integrating by parts moves the derivative off ϕ and onto j^μ ; the boundary term is discarded as usual; and what remains, $\frac{1}{M}\phi(\partial_\mu j^\mu)$, is $\frac{1}{M} \times 0$ if the current is conserved. A divergent coefficient multiplying zero: the dangerous coupling simply is not there.

Why this is more than a technical trick. What has been shown is that the massless limit exists *if and only if* the vector couples to a conserved current — which is a strong, physical statement. It is the Lagrangian version of the Ward identity of Lecture 11: there, the redundancy $\varepsilon^\mu \rightarrow \varepsilon^\mu + Cp^\mu$ forced $p_\mu \mathcal{M}^\mu = 0$; here, the requirement that the longitudinal mode decouple forces $\partial_\mu j^\mu = 0$. Same physics, two languages.

A caveat worth knowing (not in the notes). The smoothness established here is a *classical, tree-level* statement for a vector coupled to a conserved current. The analogous limit for a massive *graviton* famously fails — the van Dam–Veltman–Zakharov discontinuity — because the extra polarizations of the spin-2 field couple to the trace of the energy–momentum tensor and do not decouple. The spin-1 case is the benign one.

9 Consistency with gauge invariance [p. 35*****–36*****]

Obs.: This result is consistent with the issue of gauge invariance. We know that a successful description of a massless spin-1 field needs a theory that is invariant if we transform $V^\mu(x) \rightarrow V^\mu(x) + \partial^\mu\alpha(x)$.

↪ Let's check this with our Lagrangian:

$$\begin{aligned} \mathcal{L} = & \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\substack{\text{this term is manifestly} \\ \text{gauge-invariant if we transform} \\ V^\mu \rightarrow V^\mu + \partial^\mu\alpha}} \quad \underbrace{-V_\mu j^\mu}_{\text{this term transforms to}} \\ & -V_\mu j^\mu \longrightarrow -(V_\mu + \partial_\mu\alpha)j^\mu = -V_\mu j^\mu - (\partial_\mu\alpha)j^\mu \\ & \stackrel{\text{IBP}}{=} -V_\mu j^\mu + \alpha(\partial_\mu j^\mu) \stackrel{\partial_\mu j^\mu=0}{=} -V_\mu j^\mu \end{aligned}$$

And it is invariant since the current is conserved.

Example: consider the theory

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu + \bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi - gV_\mu\bar{\Psi}\gamma^\mu\Psi - g_5V_\mu\bar{\Psi}\gamma^\mu\gamma_5\Psi$$

The “naive” massless limit is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi - V_\mu(g\bar{\Psi}\gamma^\mu\Psi + g_5\bar{\Psi}\gamma^\mu\gamma_5\Psi)$$

Added explanation (not in the original notes)

The same integration by parts, twice. Notice that the calculation here is character for character the one on [p. 35*****] — move the derivative off the gauge parameter and onto the current, then use conservation. That is not a coincidence: the massless limit and gauge invariance are the *same* requirement seen from two sides, which is exactly the “consistency” the notes point to.

Why $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is manifestly invariant. Under $V^\mu \rightarrow V^\mu + \partial^\mu\alpha$,

$$F^{\mu\nu} \rightarrow F^{\mu\nu} + \partial^\mu\partial^\nu\alpha - \partial^\nu\partial^\mu\alpha = F^{\mu\nu},$$

the added terms cancelling by symmetry of mixed partials — the same cancellation seen on [p. 27*****]. The field strength is gauge invariant on the nose, no integration by parts required.

Where the mass term stands. $\frac{1}{2}M^2V_\mu V^\mu$ transforms into $\frac{1}{2}M^2(V + \partial\alpha)^2 \neq \frac{1}{2}M^2V^2$ — not invariant, and no integration by parts rescues it. This is why gauge invariance appears only in the massless limit and why $M \neq 0$ and gauge invariance are incompatible for a single vector field. (The Higgs mechanism evades this by restoring the eaten scalar — the Stückelberg ϕ of [p. 27*****].)

10 Is the current conserved? The axial obstruction [p. 37*****–38*****]

Is the current $j^\mu \equiv g\bar{\Psi}\gamma^\mu\Psi + g_5\bar{\Psi}\gamma^\mu\gamma_5\Psi$ conserved?

The theory admits a global internal $U(1)$ symmetry under which

$$\begin{cases} \Psi(x) \rightarrow e^{-iq\theta}\Psi(x) \\ \bar{\Psi}(x) \rightarrow e^{+iq\theta}\bar{\Psi}(x) \end{cases}$$

Let’s compute the corresponding Noether current. The formula is

$$j^\mu = \underbrace{\frac{\partial\mathcal{L}}{\partial(\partial_\mu\Psi)}}_{= i\bar{\Psi}\gamma^\mu} D\Psi$$

From the infinitesimal variation

$\Psi(x) \rightarrow \Psi'(x) = e^{-iq\theta}\Psi(x) = \Psi(x) - iq\theta\Psi(x)$, so that $D\Psi = -iq\Psi(x)$.

So that we get

$$j^\mu = q\bar{\Psi}\gamma^\mu\Psi$$

△ Consequently, since the part $g_5\bar{\Psi}\gamma^\mu\gamma_5\Psi$ is not conserved, the theory does not admit a sensible massless limit.

However, if we restrict to a theory with Lagrangian density

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}M^2V_\mu V^\mu + \bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi - \underbrace{gV_\mu\bar{\Psi}\gamma^\mu\Psi}$$

It does admit a sensible massless limit since the coupling occurs with a conserved current!

$$\downarrow \quad M \rightarrow 0$$

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\Psi}(i\gamma^\mu\partial_\mu - m)\Psi - qA_\mu\bar{\Psi}\gamma^\mu\Psi$$

This is the ordinary QED Lagrangian.

The charge “ q ” plays naturally the role of coupling constant.

Added explanation (not in the original notes)

Why the axial current is not conserved. The Dirac Lagrangian is invariant under the *vector* rotation $\Psi \rightarrow e^{-iq\theta}\Psi$ (both $\bar{\Psi}\Psi$ and $\bar{\Psi}\gamma^\mu\partial_\mu\Psi$ are unchanged), which is why Noether’s theorem returns $q\bar{\Psi}\gamma^\mu\Psi$. The *axial* rotation $\Psi \rightarrow e^{-i\theta\gamma_5}\Psi$ instead flips the sign of the mass term, because γ_5 anticommutes with γ^μ :

$$\partial_\mu(\bar{\Psi}\gamma^\mu\gamma_5\Psi) = 2im\bar{\Psi}\gamma_5\Psi \neq 0 \quad \text{for } m \neq 0.$$

So the axial current is conserved only for massless fermions — and, famously, not even then once quantum corrections are included (the ABJ anomaly).

What this rules in and out. A massive vector coupled purely to a *vector* current has a smooth massless limit: that limit is QED. A massive vector with an axial coupling does not. This is not an academic point — it is why the W and Z bosons of the weak interaction, which do couple axially, cannot simply be given a mass by hand: one needs the Higgs mechanism, where the would-be longitudinal mode is supplied by a genuine scalar field rather than being an artifact of the decomposition.

Transcription note. On [p. 38*****] the coupling in the massless Lagrangian is written as q while the massive Lagrangian above it uses g ; the notes remark explicitly that “the charge q plays naturally the role of coupling constant”, so the two symbols denote the same quantity once the current is identified as the Noether current of the $U(1)$. The field is also renamed $V^\mu \rightarrow A^\mu$ at that point, as befits a photon.

11 The photon propagator as a massless limit [p. 38*****–40*****]

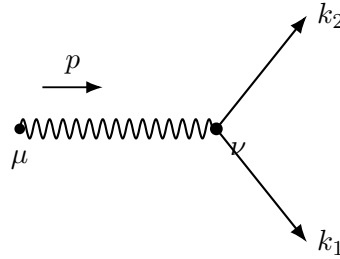
Comment

Let us try to derive the photon propagator as the massless limit of a massive one:

$$\mu \overset{p}{\longrightarrow} \bullet \text{~~~~~} \bullet \nu \equiv \frac{i}{p^2 - M^2 + i\epsilon} \left(-g^{\mu\nu} + \frac{p^\mu p^\nu}{M^2} \right)$$

$\hookrightarrow$ The limit $M \rightarrow 0$ seems to diverge...

However, if the coupling occurs with a conserved current, these “dangerous” terms do not contribute. Let us illustrate this point with an example.



$$= \frac{i}{p^2 - M^2 + i\epsilon} \left(-g^{\mu\nu} + \frac{p^\mu p^\nu}{M^2} \right) \times \left[\bar{u}(k_2, \lambda_2) (-iq\gamma^\mu) u(k_1, \lambda_1) \right]$$

$$= \underbrace{\frac{(-i)g^{\mu\nu}}{p^2 - M^2 + i\epsilon} \left[\bar{u}(k_2, \lambda_2) (-iq\gamma^\mu) u(k_1, \lambda_1) \right]}_{\text{the massless limit is harmless in this term}}$$

$$+ \frac{i(p^\mu p^\nu / M^2)(-iq)}{p^2 - M^2 + i\epsilon} \underbrace{\left[\bar{u}(k_2, \lambda_2) \not{p} u(k_1, \lambda_1) \right]}_{\text{we have } k_1 + p = k_2}$$

$$\bar{u}(k_2, \lambda_2) (\not{k}_2 - \not{k}_1) u(k_1, \lambda_1) = \bar{u}(k_2, \lambda_2) (m - m) u(k_1, \lambda_1) = 0$$

applying the Dirac equation

The dangerous term, therefore, vanishes. We thus write for the photon propagator

$$\begin{array}{c} \mu \quad \quad \quad \nu \\ \bullet \text{---} \text{wavy line} \text{---} \bullet \\ \quad \quad \quad \xrightarrow{p} \end{array} \equiv \frac{-i g^{\mu\nu}}{p^2 + i\epsilon}$$

Photon propagator (in the so-called Feynman gauge)

Added explanation (not in the original notes)

Transcription note on indices. The propagator carries the index pair $\mu\nu$ while the vertex is written γ^μ ; strictly, one of the two indices must be contracted, so the vertex factor should read $-iq\gamma^\nu$ (or the propagator index be relabelled). The notes are being schematic — and the point of the calculation, which index sits where notwithstanding, is that the $p^\mu p^\nu$ term contracts a momentum into the fermion line.

The Dirac-equation step, in detail. Momentum conservation at the vertex gives $p = k_2 - k_1$, so $\not{p} = \not{k}_2 - \not{k}_1$. Now use the Dirac equations for the external spinors:

$$\bar{u}(k_2) \not{k}_2 = m \bar{u}(k_2), \quad \not{k}_1 u(k_1) = m u(k_1),$$

so the sandwich gives $\bar{u}(k_2)(m - m)u(k_1) = 0$ exactly. Nothing is neglected — the term is identically zero, not merely small.

This is the Ward identity again. $p_\mu \mathcal{M}^\mu = 0$ is precisely what makes the $p^\mu p^\nu$ piece drop. Lecture 11 derived that identity from the redundancy of polarization vectors; here it is verified by explicit computation on a concrete diagram. Note that the mechanism is *on-shell external legs plus current conservation* — the same two ingredients as in the Lagrangian argument on [p. 35*****].

Why “Feynman gauge”. The result $-ig^{\mu\nu}/(p^2 + i\epsilon)$ is the photon propagator in a particular gauge. The general form is

$$\frac{-i}{p^2 + i\epsilon} \left[g^{\mu\nu} - (1 - \xi) \frac{p^\mu p^\nu}{p^2} \right],$$

with $\xi = 1$ giving Feynman gauge and $\xi = 0$ Landau gauge. The gauge-dependent $p^\mu p^\nu$ piece never contributes to physical amplitudes — for exactly the reason just demonstrated — which is why one is free to choose whichever ξ makes a calculation easiest.

Why this whole detour was necessary. A massless gauge field cannot be quantized naively: the Maxwell kinetic operator is not invertible (it annihilates pure-gauge configurations), so “the propagator” does not exist until a gauge is fixed. Obtaining it as the $M \rightarrow 0$ limit of the Proca propagator, as done here, sidesteps gauge fixing entirely — the massive theory has no gauge freedom and its propagator is perfectly well defined. The limit is legitimate precisely because the coupling is to a conserved current.